

Airframe Technology Master

Certification

Lecture 04

Structural Requirements

Paola Martín Gallego

Paz Pastor Molina

April 2025

STRUCTURAL REQUIREMENTS (II)

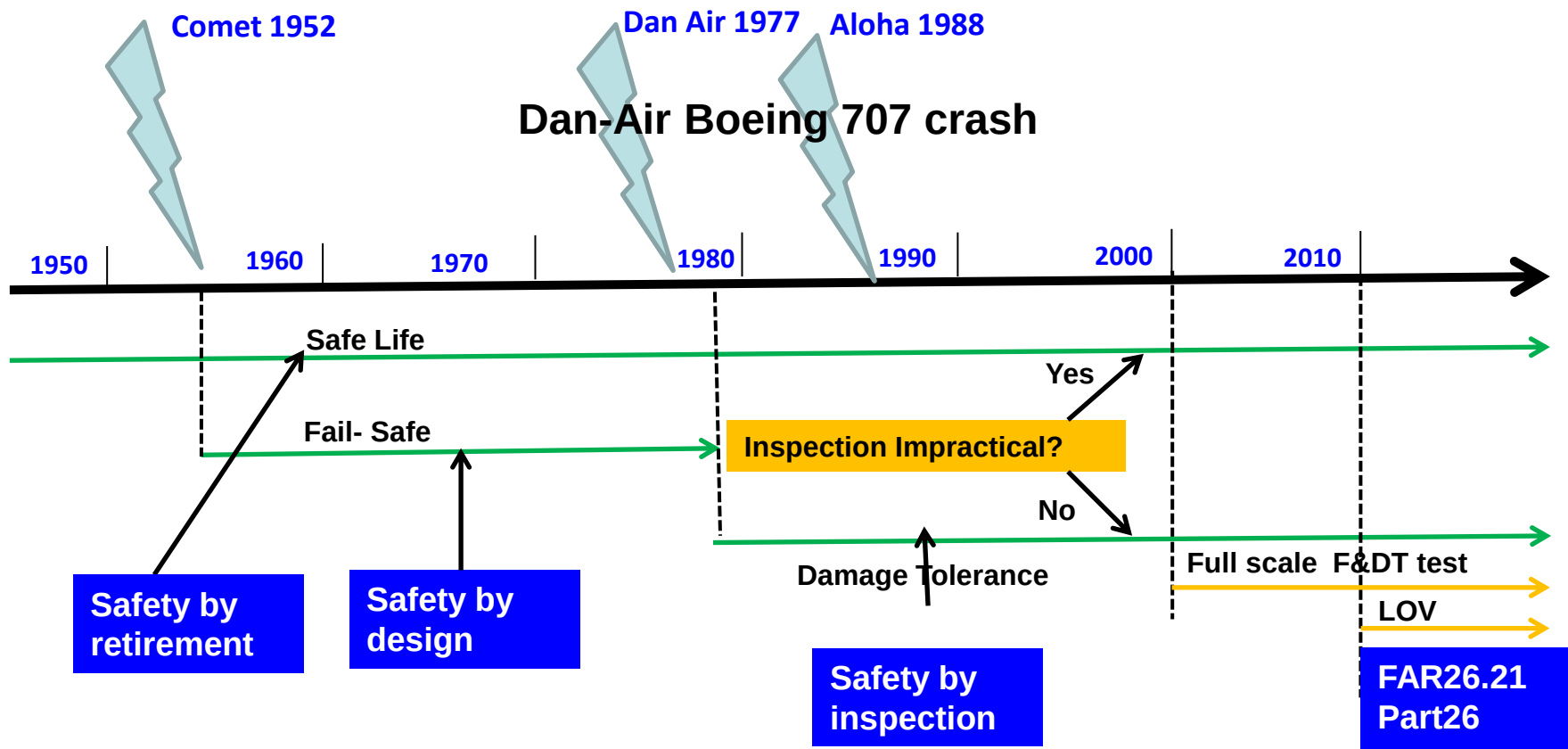
AGENDA

Applicable structural requirements

- Fatigue and Damage Tolerance
- Impact Resistance
- Crashworthiness
- Ditching
- Rapid Decompression
- Windshields & Windows
- Aeroelasticity
- Interaction of systems and structures
- Instructions for Continued Airworthiness, ALS (CS 25.1529)

Evolution of Fatigue Requirements

Certification requirements have changed over the years, largely based on key historical events.



De Havilland D. H. 106 “Comet 1”, the first jet airliner to enter in service (BOAC, 2nd May of 1952)

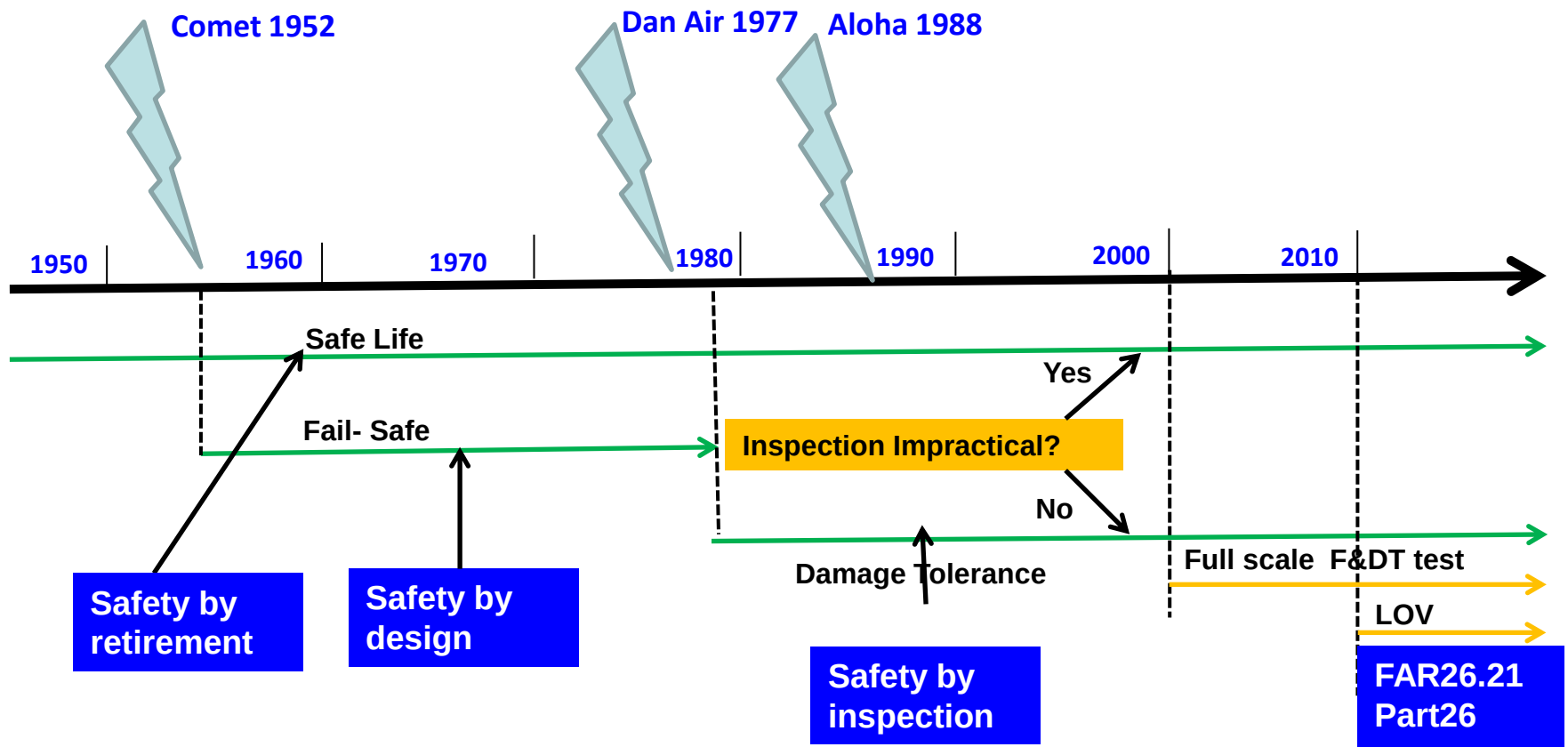
- Several aircraft crashed: After one year of operation first one crashed at Calcutta. The next year, two aircrafts more disintegrated while were flying at 35000 ft.
- As a consequence, airworthiness certificate was removed and a formal investigation was launched. A fatigue test was initiated in a old aircraft with 1230 FC and at 1830 FC the failure occurs. Analysis of this failure together with the analysis of the debris of one aircraft disintegrated revealed that was due to a high stress concentration factor at windows corner.
- The Comet fuselage was fatigue tested without finding any cracks similar to the ones appeared in service.
- The cause was to use the same test specimen for static and fatigue test.
- De Havilland's fatigue test setup did not reflect the production fleet. The test specimen had been previously cold worked due to overloaded proof pressure testing. This overload inadvertently enhanced the fatigue life of the specimen, and thus produced results which did not reflect the production configuration. Today is well known that the application of a static load cause a sensible improve in the fatigue behaviour.

**FAIL-SAFE
CONCEPT**

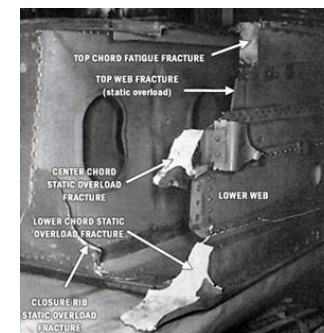


Evolution of Fatigue Requirements

Certification requirements have changed over the years, largely based on key historical events.

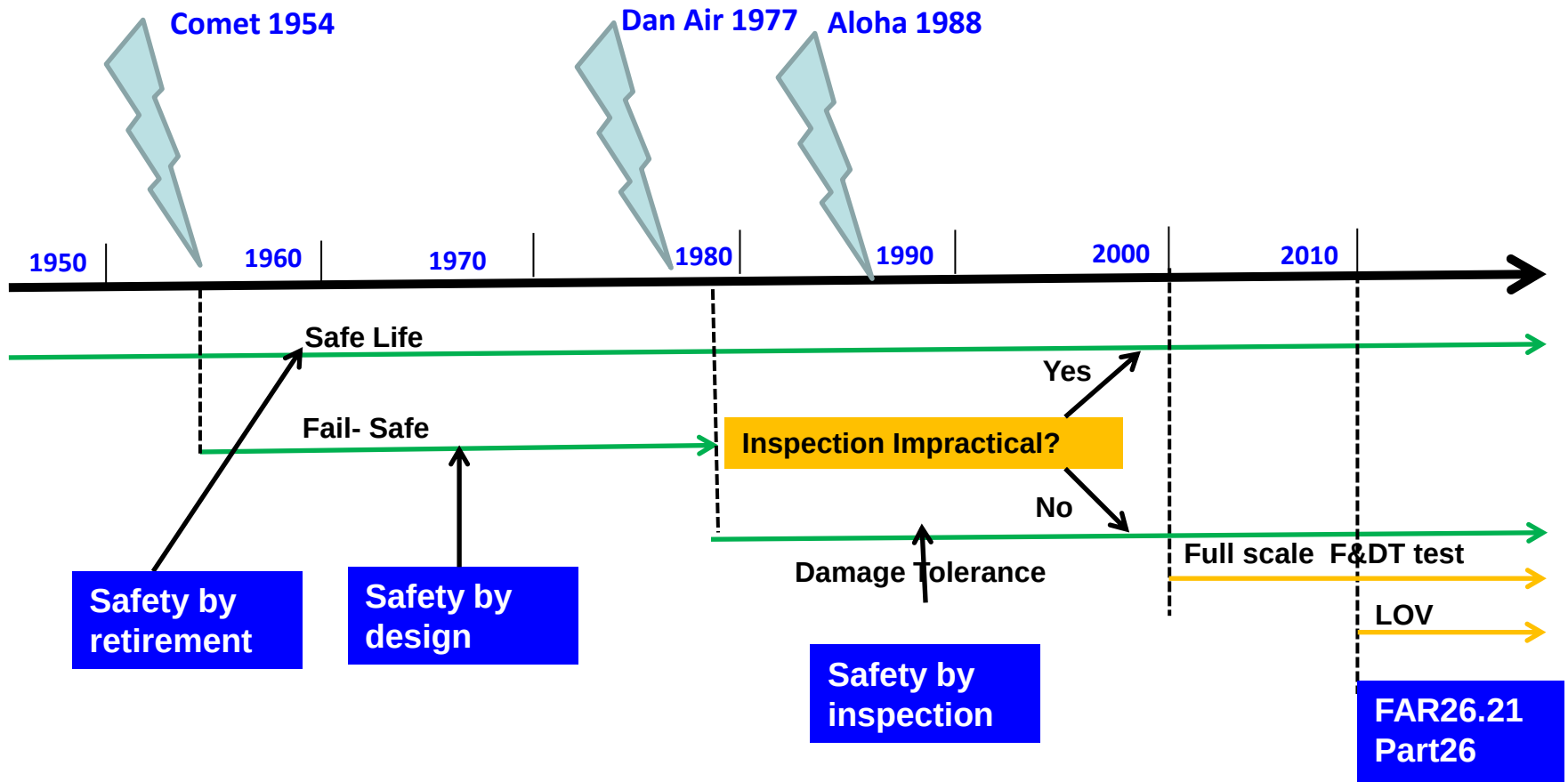


- B707-300 1977 at Lusaka airport crash.
 - The aircraft crashed during approach. All six occupants of the aircraft were killed.
 - During the descend phase, witnesses saw the entire right horizontal stabiliser and elevator assembly detach from the aircraft. The aircraft subsequently lost pitch control and entered a nose-dive from about 800 ft (240 m) to ground level, destroying the aircraft on impact.
 - The 707 had been designed to a fail-safe philosophy, and failure of the tailplane rear spar had been calculated to be insufficient to cause the loss of the aircraft, the remaining front spar being sufficiently strong to enable the aircraft to land safely. The fail-safe philosophy of the 707-100 was supported by test. When the model 707-300 was developed, the stabilizer assembly was extensively redesigned and after the investigation it was concluded the test of 707-100 was not representative for 707-300.
 - The load increase on the stabilizer that occurred as a result of the stiffening of the skin panel led to the development of fatigue cracking in an area outboard of the lug, new feature design of 707-300 versus 707-100. This outboard location, which was not considered to be a likely failure scenario, was not analyzed by Boeing.



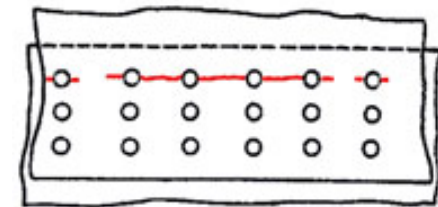
Evolution of Fatigue Requirements

Certification requirements have changed over the years, largely based on key historical events.



- B737 1988 at Hawaii - Aloha accident : Loss of upper part of fuselage flying at 24000ft.
 - Aircraft was able to land with all passengers except one crew member who was expelled to exterior at the accident time.
 - Investigations showed that in the lap-joints the bonding had been degraded by the hot, humid and salty environment. This led to a higher loading in the riveting which created a number of small cracks that in the end explosively grow together.
 - Root cause was multiple crack coalescence at a longitudinal joint that cause such a big crack with explosive results for the normal pressurization load case.
 - This phenomenon called widespread fatigue damage led to another change in regulations, namely CS 25.571(b). Today damage tolerance is part of the Aircraft development and justification.

WIDESPREAD FATIGUE DAMAGE



Fatigue and Damage Tolerance CS 25.571

- (a) General. An evaluation of the strength, detail design, and fabrication must show that **catastrophic failure** due to **fatigue, manufacturing defects, environmental deterioration, or accidental damage**, will be avoided throughout the **operational life**...

...An **analysis, supported by test evidence, of the principal structural elements** and detail design points identified...

→ Maintain the structural integrity over the whole aircraft operational life

- Two different aspects

Durability  Economy Safety  Airworthiness

- Basic Phenomena

Fatigue, manufacturing defects, environmental deterioration, Accidental Damage

Fatigue and Damage Tolerance CS 25.571 (Cont'd)

Some definitions:

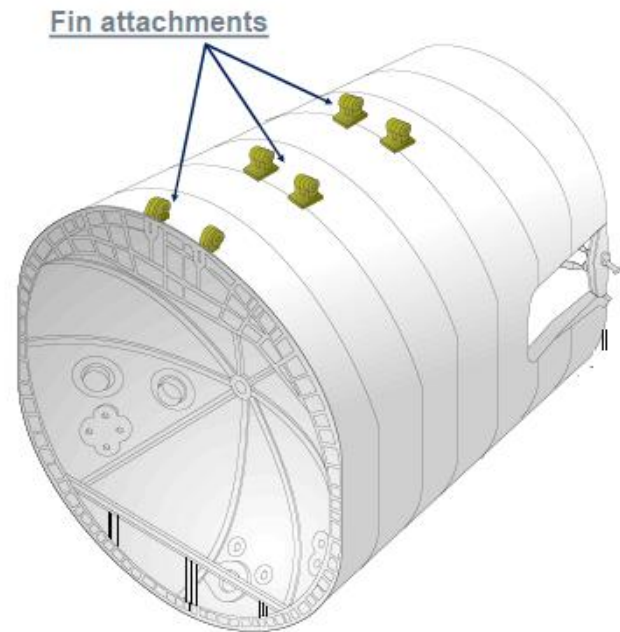
PRINCIPAL STRUCTURAL ELEMENTS (PSE): are those which contribute significantly to carrying flight, ground or pressurisation loads, and whose failure if it remains undetected could result in catastrophic failure of the aeroplane.

WING

- Control surfaces, slats, flaps,
- Primary fittings

FUSELAGE

- Frames
- Skin around cut-out
- Door frame
- Pressure bulkhead



CS 25.571 is only formally applicable when a component is PSE

Fatigue and Damage Tolerance CS 25.571

For the following types of structure, the threshold **must be established based on analyses and/or tests**, assuming the structure contains an initial flaw representative of a defect or damage of the maximum probable size that could exist as a result of manufacturing processes or manufacturing or service induced damage:

- (i) **single load path structure; and**
- (ii) **multiple load path 'fail-safe' structure and crack arrest 'fail-safe' structure, where it cannot be demonstrated that the resulting load path failure or partial failure (including arrested cracks) will be detected and repaired during normal maintenance, inspection, or operation of an aeroplane prior to failure of the remaining structure.**

Based on the evaluations required by this paragraph, inspections or other procedures must be established, as necessary, to prevent catastrophic failure, and must be included in the Airworthiness Limitations Section of the Instructions for Continued Airworthiness required by CS 25.1529. **The limit of validity of the engineering data that supports the structural maintenance programme** (hereafter referred to as **LOV**), stated as a number of total accumulated flight cycles or flight hours or both, established by this paragraph, must also be included in the Airworthiness Limitations Section of the Instructions for Continued Airworthiness

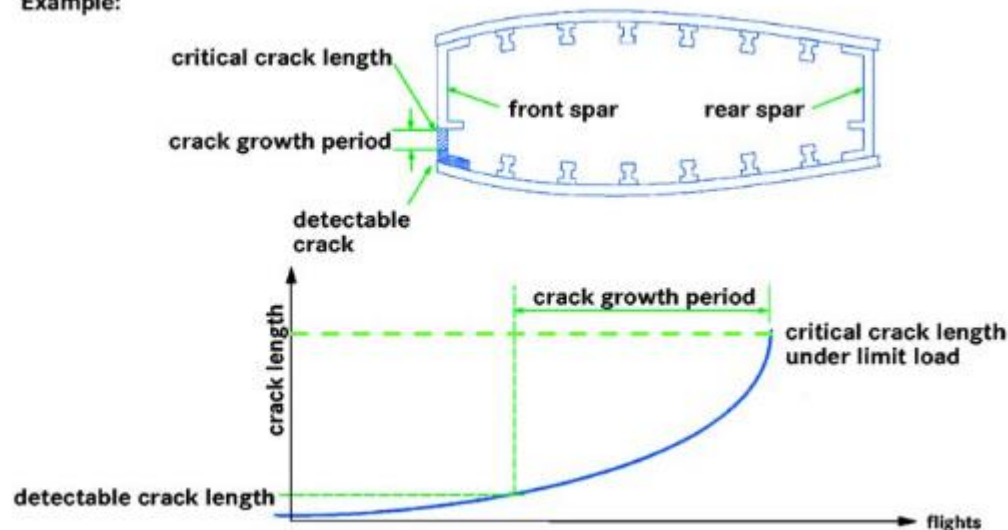


Fatigue and Damage Tolerance CS 25.571 (Cont'd)

Some definitions:

SINGLE LOAD-PATH: is where applied loads are eventually distributed through a single member within an assembly, the failure of which would result in the loss of the structural integrity of the component involved.

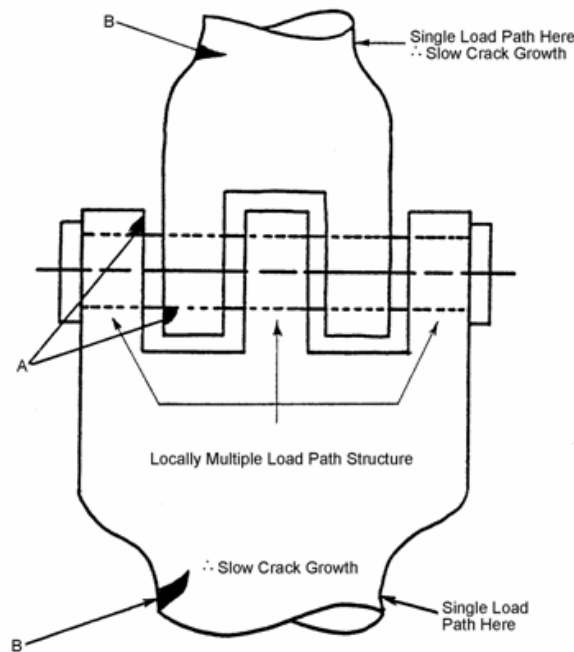
Example:



Fatigue and Damage Tolerance CS 25.571 (Cont'd)

Some definitions:

MULTIPLE LOAD PATH: is identified with redundant structures in which (with the failure of individual elements) the applied loads would be safely distributed to other load carrying members.



Fatigue and Damage Tolerance CS 25.571 (Cont'd)

Some definitions:

FAIL-SAFE: means that the structure has been evaluated to assure that catastrophic failure is not probable after fatigue failure or obvious partial failure of a single principal structural element. In case of failure of one part, the surrounding structure is used as a secondary load path that must be capable of taking over the load carrying capability of the failed part.

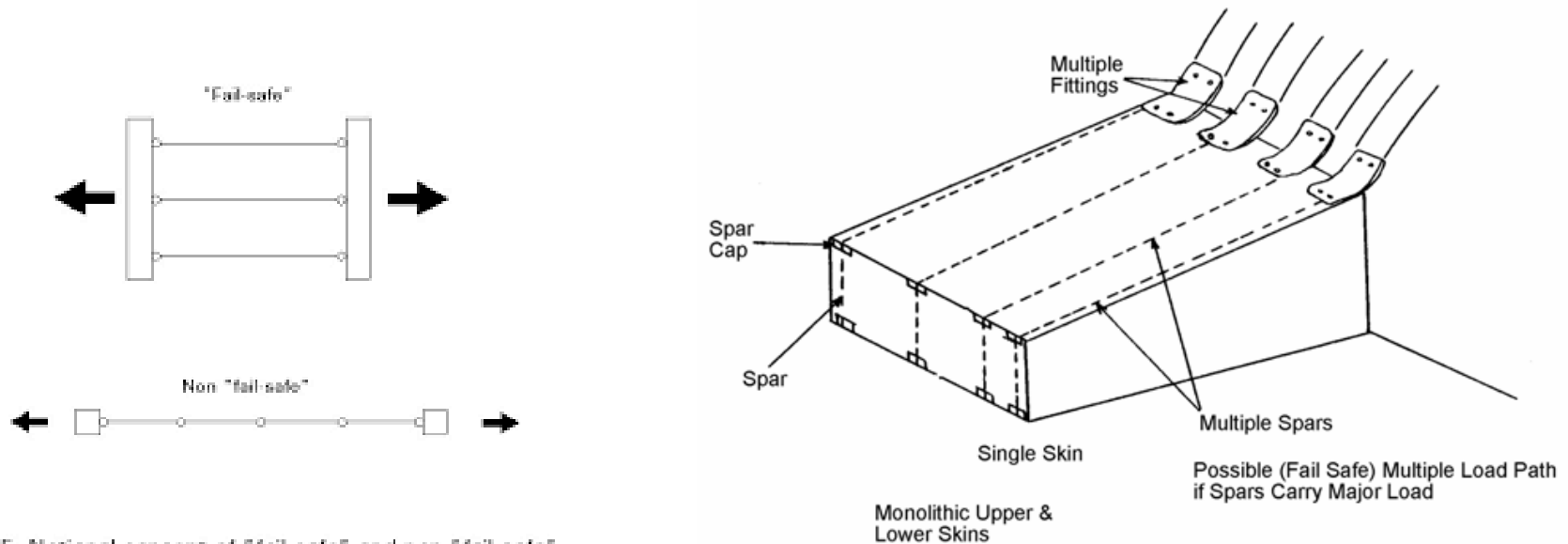


Figure 5 Notional concept of "fail-safe" and non "fail-safe" structural elements

Fatigue and Damage Tolerance CS 25.571

(b) The evaluation must include a determination of the **probable location** and **modes of damages** due to fatigue, environmental deterioration (e.g corrosion) or accidental damage...

Repeated load and static analyses, supported by test evidence and (if available) service experience, must be incorporated in the evaluation. Damage at multiple sites due to prior fatigue exposure (including special consideration of widespread fatigue damage) must be included in the evaluation where the design is such that this type of damage could occur. An **LOV** must be established that corresponds to the period of time, stated as a number of total accumulated flight cycles or flight hours or both, for which it has been demonstrated by full-scale fatigue test evidence that **widespread** fatigue damage will not occur in the aeroplane structure.

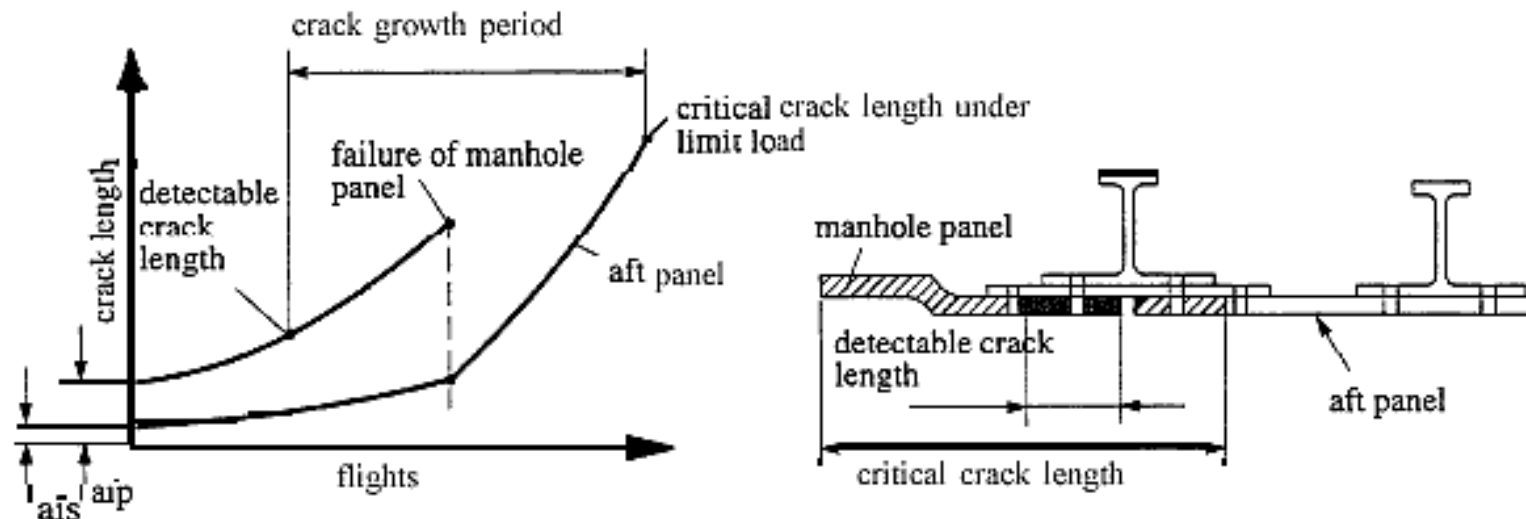
The extent of damage for residual strength evaluation at any time within the operational life must be **consistent with the initial detectability and subsequent growth under repeated loads. The residual strength evaluation must show that the remaining structure is able to withstand loads (considered as static ultimate loads)** corresponding to ... Limit symmetrical manoeuvring, gust, rolling, yaw manoeuvring, pressurisation and ground loads.

- Fatigue & damage tolerance analyses allow definition of safe inspection program.
- Drop of residual strength below ultimate load within inspection interval is acceptable

Fatigue and Damage Tolerance CS 25.571 (Cont'd)

Some definitions:

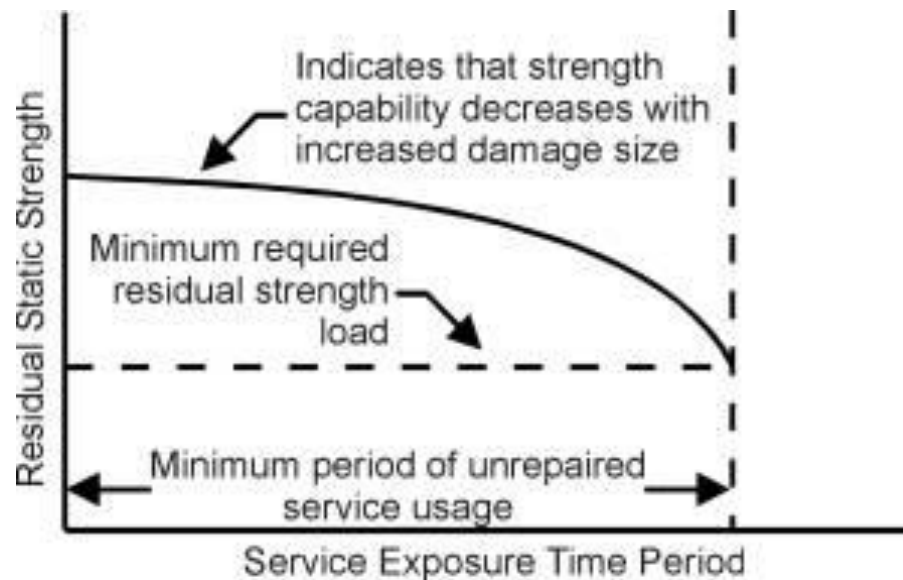
DAMAGE TOLERANCE: means that the structure has been evaluated to ensure that should serious fatigue, corrosion or accidental damage occur within the operational life of the airplane, the remaining structure can withstand the loads without failure or excessive structural deformation until the damage is detected.



Fatigue and Damage Tolerance CS 25.571 (Cont'd)

Some definitions:

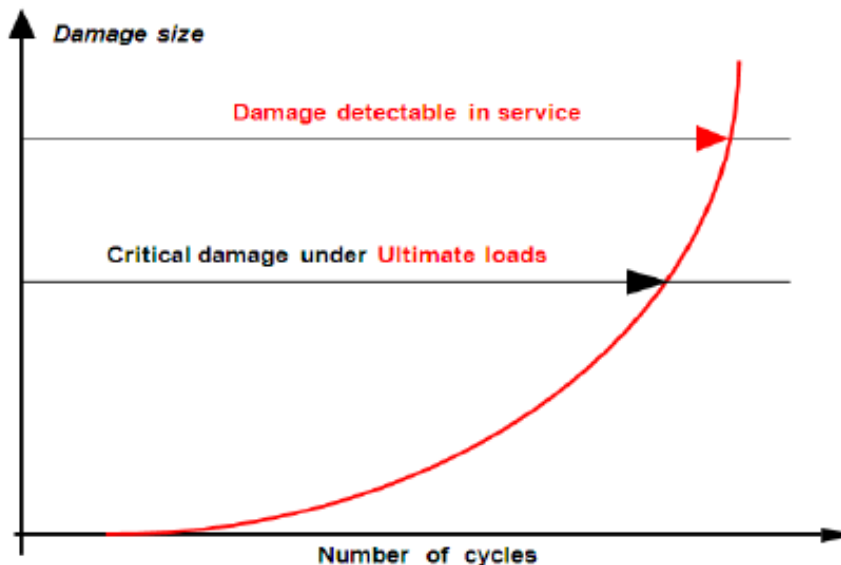
RESIDUAL STRENGTH: means the strength of the structure in presence of damage.



Fatigue and Damage Tolerance CS 25.571 (Cont'd)

(c) Fatigue (**safe-life**) evaluation. Compliance with the damage-tolerance requirements of subparagraph (b) of this paragraph is not required if the applicant establishes that their application for particular structure is impractical. **This structure must be shown by analysis, supported by test evidence, to be able to withstand the repeated loads of variable magnitude expected during its service life without detectable cracks.**

Appropriate safe-life scatter factors must be applied.

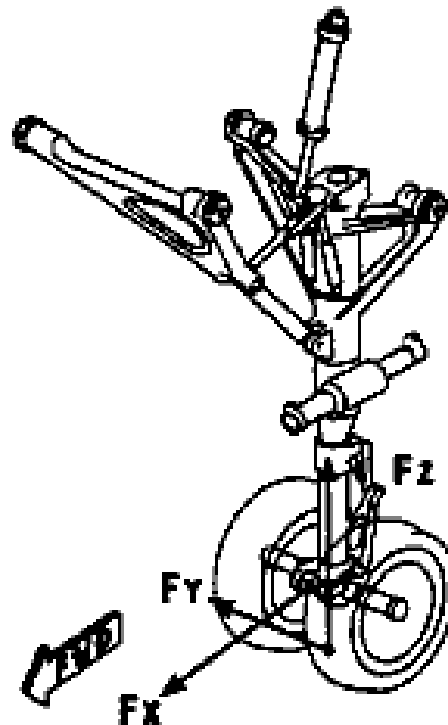


Life Limit=Tested Life/Scatter Factor

Fatigue and Damage Tolerance CS 25.571 (Cont'd)

Some definitions:

SAFE-LIFE: means that the structure has been evaluated to be able to withstand the repeated loads of variable magnitude expected during its service life **without detectable cracks.**



Fatigue and Damage Tolerance CS 25.571 (Cont'd)

(d) Sonic fatigue strength. It must be shown by analysis, supported by test evidence, or by the service history of aeroplanes of similar structural design and sonic excitation environment, that –

(1) **Sonic fatigue cracks are not probable** in any part of the flight structure subject to sonic excitation; or

(2) **Catastrophic failure caused by sonic cracks is not probable** assuming that the loads prescribed in sub-paragraph (b) of this paragraph are applied to all areas affected by those cracks.

Usual Means of Compliance used

0, 1, 2, 4, 6

Fatigue and Damage Tolerance CS 25.571 (Cont'd)

(e) Damage-tolerance (discrete source) evaluation. The aeroplane must be capable of successfully completing a flight during which likely structural damage occurs as a result of **bird impact** as specified in CS25.631;

The damaged structure must be able to withstand the static loads (considered as ultimate loads) which are reasonably expected to occur at the time of the occurrence and during the completion of the flight [...]

- Justification are made by analyses and /or by tests.
- This test are mainly performed to prove the durability (determination of the life limits), to identify or confirm the critical sites, i.e, hot spot, and to validate the analysis.
- This test could be partial (component test) or Full Scale.
- Early in the programme, the finding of a damage on a tested specimen provides more opportunities for corrective actions. And for the in-service fleet, the test provides a better knowledge of loads and structure.

Impact Resistance .-bird strike

CS 25.571 (e) refers to CS25.631 for bird impact conditions while FAR 25.571 (e) includes them, dedicating FAR 25.631 to a more restrictive requirement, just for empennage:

CS 25.631 Bird Strike Damage

The aeroplane must be designed to assure capability of **continued safe flight and landing** of the aeroplane after impact with a **4 lb bird** when the velocity of the aeroplane (relative to the bird along the aeroplane's flight path) is equal to **VC at sea level or 0,85 VC at 2438m (8000 ft.)**, whichever is the most critical.

Compliance may be shown by analysis only when based on tests carried out on sufficiently representative structures of similar design (see AMC 25.631).

Impact Resistance .-bird strike

FAR 25.631 Bird Strike Damage (specifity)

The empennage structure must be designed to assure capability of continued safe flight and landing of the airplane after impact with an **8-pound bird** when velocity of the airplane (relative to the bird along the aeroplane's flight path) is equal to VC at sea level, selected under paragraph 25.335(a)). Compliance with this section by provision of redundant structure and protected location of control system elements or protective devices such as splitter plates or energy absorbing material is acceptable. Where compliance is shown by analysis, tests, or both, use of data of airplanes having similar structural design is acceptable.



Impact Resistance includes also following CS/FAR:

CS 25.734 Protection against wheel and tyre failures

CS 25.775 Windshields and Windows

CS 25.903 Uncontained Engine Failure

CS 25.905 Propellers

CS 25.963 Fuel Tanks



Crashworthiness : Emergency landing conditions CS 25.561

(a) The aeroplane, although it may be damaged in emergency landing conditions on land or water, must be designed as prescribed in this paragraph to protect each occupant under those conditions.

(b) **The structure must be designed** to give each occupant every reasonable chance of escaping serious injury in a minor crash landing when –

(1) Proper use is made of seats, belts, and all other safety design provisions;

(2) The wheels are retracted (where applicable); and

(3) The occupant experiences the following **ultimate inertia forces acting separately** relative to the surrounding structure:

(i) Upward, 3·0g (ii) **Forward, 9·0g**

(iii) Sideward, 3·0g on the airframe and 4·0g on the seats and their attachments

(iv) Downward, 6·0g (v) Rearward, 1·5g [...]

Crashworthiness: Emergency landing conditions CS 25.561 (Cont'd)

(c) **For equipment, cargo in the passenger compartments and any other large masses,** the following apply:

(1) These items must be positioned so that if they break loose they will be unlikely to:

(i) Cause direct injury to occupants;

(ii) Penetrate fuel tanks or lines or cause fire or explosion hazard by damage to adjacent systems; or (iii) nullify any of the escape facilities provided for use after an emergency landing.

(2) **When such positioning is not practical** (e.g. fuselage mounted engines or auxiliary power units) **each such item of mass must be restrained under all loads up to those specified in subparagraph (b)(3) of this paragraph.** The local attachments for these items should be designed to withstand **1.33 times** the specified loads if these items are subject to **severe wear and tear** through frequent removal (e.g. quick change interior items).

(d) Seats and items of mass (and their supporting structure) must not deform under any loads up to those specified in sub-paragraph (b)(3) of this paragraph in any manner that would impede subsequent rapid evacuation of occupants. (See AMC 25.561(d).)

Also related paragraphs CS 25.787 & CS 25.789

Crashworthiness: Emergency landing conditions CS 25.561 (Cont'd)

The aim of Crashworthiness is to make the aircraft as safe as can reasonably be expected under otherwise survivable crash conditions. It must be ensured that the overall aircraft design meets the applicable crashworthiness regulations such that in the unlikely event of an emergency landing on land.



Emergency Landing in Water are included in following CS:

CS 25.561 Ditching Requirements

CS 25.563 Structural Ditching Provisions

CS 25.801 Ditching

CS 25.807 (i) Flotation Time

Approach



- Pitch
- Vertical Speed

Impact



- Pressure distribution
- Local deformation

Landing



- Risk of nose diving
- Global deformation

Floatation

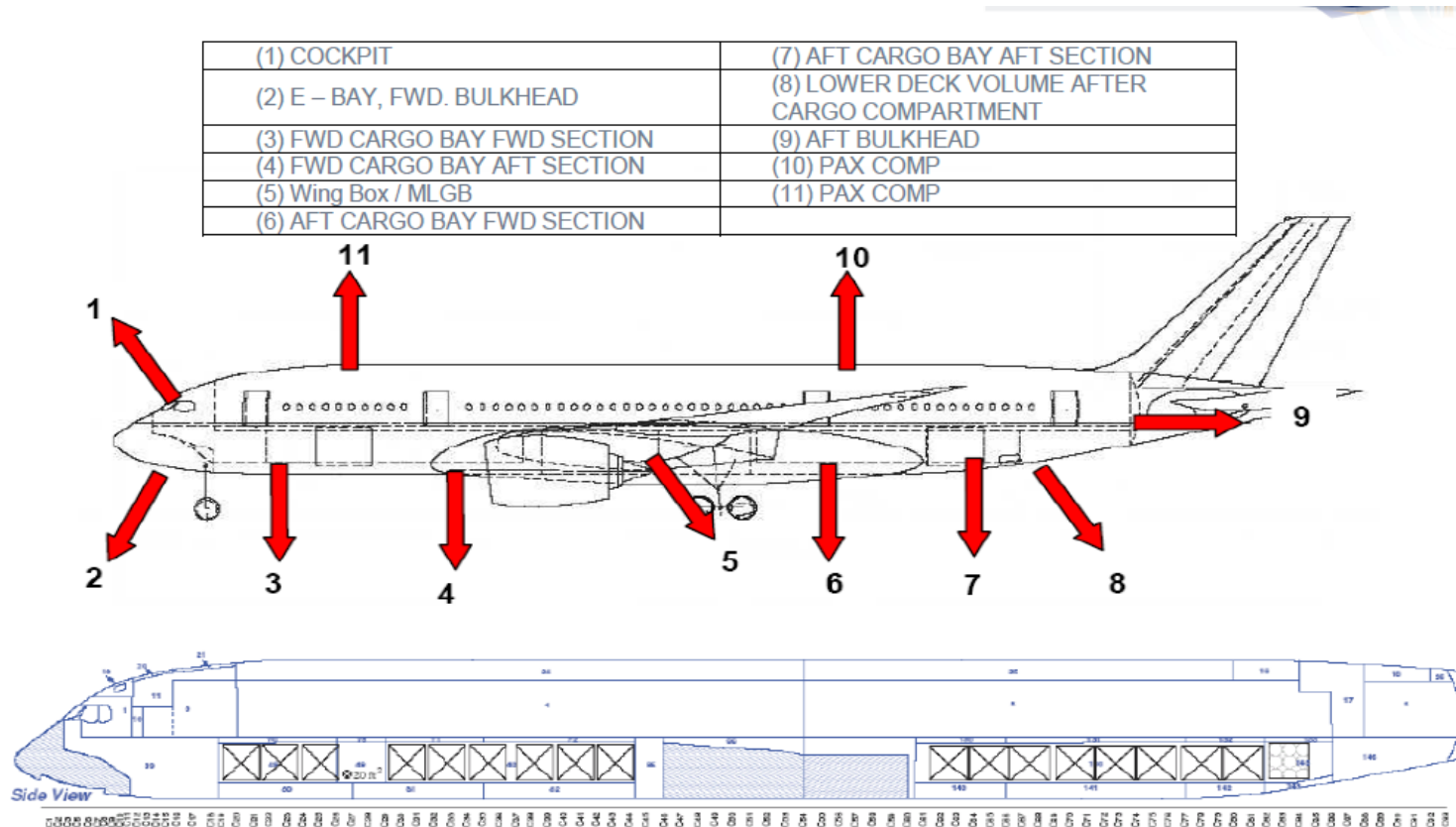


- Floatation time
- Evacuation

It will be checked that the fuselage can withstand this loads without failure and that the surrounding structure of the emergency exit door to be used after ditching will not be deformed in such a way that the door cannot be open.



Structure must be designed to withstand the effects of a sudden release of pressure through an opening in any compartment (CS25.365 (e) and (f))



Loss of fan cowl

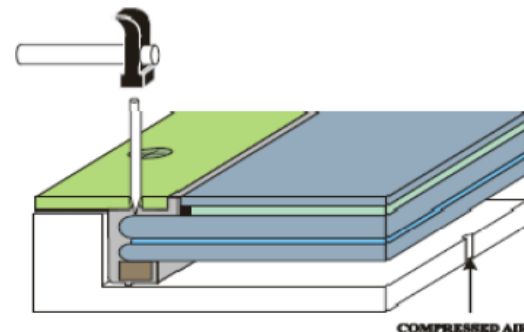


Windshields & Windows

CS 25.775 - Windshields & Windows

(b) Windshield panes directly in front of the pilots in the normal conduct of their duties, and the supporting structures for these panes, must withstand, without penetration, the **bird impact** conditions specified in CS 25.631.

(d) The design of windshields and windows in pressurised aeroplanes must be based on factors peculiar to high altitude operation, including the effects of continuous and **cyclic pressurisation** loadings, the inherent characteristics of the material used, and the effects of temperatures and temperature differentials. The windshield and window panels must be capable of **withstanding the maximum cabin pressure differential loads** combined with critical aerodynamic pressure and temperature effects **after any single failure** in the installation or associated systems.



Aeroelasticity (CS25.629)

The Aeroelasticity stability evaluation include **flutter, divergence, control reversal** and any undue loss of stability and control as a result of structural deformation.

Demostration of compliance with the requirements could be based of:

- Theoretical calculations before flight test
- Ground Vibration Test (GVT)
- Flight Vibration Test
- For flutter justification: calculations
- Analysis and Statement for reversal and divergence justification



Interaction of systems and Structures CS 25.302

For aeroplanes equipped with systems that affect structural performance, either directly or as a result of a failure or malfunction, the influence of these systems and their failure conditions must be taken into account when showing compliance.

These automatic system may be inoperative or may operate in a degraded mode.

It is necessary to determine the required structural factors of safety and operating margin such that the combined probability of structural failures due to application of loads during system malfunctions ensure an acceptable level of safety.

The required structural factors would be defined as a function of system reliability.

Application for systems as: Electrical flight control; Braking systems; Autopilot; Loads alleviation; Vibration alleviation; Fuel transfer

Instructions for Continued Airworthiness, ALS (CS 25.1529)

The appendix H specifies requirements for the preparation of Instructions for Continued Airworthiness as required by CS 25.1529.

H25.4 Airworthiness Limitation Section

(a) The Instructions for Continued Airworthiness must contain a section titled Airworthiness Limitations that is segregated and clearly distinguishable from the rest of the document. **This section must set forth:**

(1) Each mandatory modification time, replacement time, structural inspection interval, and related structural inspection procedure approved under CS 25.571...

...(4) A limit of validity (LOV) of the engineering data that supports the structural maintenance programme, stated as a total number of accumulated flight cycles or flight hours or both, approved under CS 25.571. Until the full-scale fatigue testing is completed and the LOV is approved, the Airworthiness Limitations Section must specify an interim limitation restricting aircraft operation to not more than **half the number of the cycles accumulated on the fatigue test article.**

(b) If the Instructions for Continued Airworthiness consist of multiple documents, the section required by this paragraph must be included in the principal manual. This section must contain a legible statement in a prominent location that reads: **‘The Airworthiness Limitations Section is approved and variations must also be approved’.**

AIRFRAME AIRWORTHINESS MATTERS

